CV

Dr. Akhilesh Kumar Maurya

Assistant Professor, Sanskriti University, Mathura
Ph.D., IIIT Allahabad | Post-Doctoral Fellow, IIT Bombay

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Citations: 350 | h-index: 10

Research Interests

Molecular Medicine; Molecular Biology; Drug Discovery, Design and Development; ADME and Toxicology; Pharmacokinetics; Cancer Biology and Signaling; Drug-Protein Interaction; Molecular Dynamics Simulation; Biomarker Identification; Basics of NGS Data Analysis; Structural Biology; In Vitro Analysis; Nanoparticle Synthesis; Nanocomposite; Drug Delivery; Cell Culture; Bacterial Culture; ELISA; MTT Assay; Biochemical Assay Analysis; Phytochemical Isolation; Plant Medicine; Phytochemicals.

Education & Research Experience

Post-Doctoral Fellow

Indian Institute of Technology Bombay, Mumbai, India
Advisor: Dr. Sushil Kumar
May 2023 – May 2025

Research: Identification of new drugs using computational drug design approaches against breast cancer target proteins and their validation on breast cancer cells.

Doctor of Philosophy (Ph.D.)

Indian Institute of Information Technology Allahabad, Prayagraj, India
Advisor: Dr. Nidhi Mishra
Jan 2018 – June 2023

Specialization: Molecular Medicine, Computational Drug Design and Development, Cancer Biology, Microbiology, Synthesis

Thesis: *CADD-Based Naturomimetic Design and Synthesis of Antimicrobial and Anti-cancerous Coumarin Derivatives and their Validation Through In-Vitro Study.*

Master of Science (Zoology)

University of Allahabad, Prayagraj, India
Dissertation Advisor: Prof. Banalata Mohanty
August 2015 – June 2017

Project: Morphology of different neurons found in *Gallus gallus* by the Golgi staining technique.

Bachelor of Science (Zoology, Chemistry)

University of Allahabad, Prayagraj, India
August 2011 – June 2014

Skills and Work Expertise

- In-silico drug designing, molecular docking (AutoDock, Maestro), pharmacophore modeling, QSAR modeling, MD simulation (GROMACS, AMBER, Desmond)
- Immunoinformatics, Microarray Data Analysis, RNA-Seq Analysis, Transcriptomics, Basics of NGS Data Analysis
- Nanoparticle synthesis and characterization; green synthesis; nanocomposite synthesis; drug delivery and controlled release
- Hands-on experience: Lyophilizer, Rotary Evaporator, Column Chromatography, TLC, HPLC, NMR data analysis, Mass data analysis
- Phytochemical isolation and activity-guided isolation
- Microbial culture, cancer cell line culture, enzymatic assay analysis, MTT assay
- Molecular cloning, transfection, Western blot, CETSA
- DNA/RNA isolation, electrophoresis, tissue and cell staining, cell line maintenance
- Linux/Windows, High-Performance Computing, MS Office
- Manuscript preparation, review, and proofreading

Patent

Patent Published Number: **202521035213**
A phosphatidylinositol 3 kinase (PI3K) alpha inhibitor for the treatment of cancer, and composition thereof.

Publications

SCI/SCIE Journals

1. Shristi Modanwal, **Akhilesh Kumar Maurya**, Mulpuru, V. et al. Exploring flavonoid derivatives as potential pancreatic lipase inhibitors for obesity management: An in silico and in vitro study. Mol Divers (2024). https://doi.org/10.1007/s11030-024-11005-5 **(IF=3.9).**
2. Shilpi Singh, **Akhilesh Kumar Maurya**, Abha Meena, Nidhi Mishra, and Suaib Luqman. Myricitrin from bayberry as a potential inhibitor of cathepsin-D: Prospects for squamous lung carcinoma prevention. *Food and Chemical Toxicology*. (2023) Vol. 179, 113988. https://doi.org/10.1016/j.fct.2023.113988. (**IF 4.3**).
3. Shilpi Singh, **Akhilesh Kumar Maurya**, Abha Meena, Nidhi Mishra, and Suaib Luqman. Myricetin 3- rhamnoside retards the proliferation of hormone-independent breast cancer cells by targeting hyaluronidase. *Journal of Biomolecular Structure and Dynamics*. (2023), https://doi.org/10.1080/07391102.2023.2256872. (**IF: 4.4**).
4. Shilpi Singh, **Akhilesh Kumar Maurya**, Abha Meena, Nidhi Mishra, and Suaib Luqman. "Narirutin downregulates lipoxygenase-5 expression and induces G0/G1 arrest in triple-negative breast carcinoma cells", *Biochimica et Biophysica Acta (BBA) - General Subjects* vol, 1867 (2023): 130340. https://doi.org/10.1016/j.bbagen.2023.130340. (**IF: 3**).
5. Shilpi Singh, **Akhilesh Kumar Maurya**, Abha Meena, Nidhi Mishra, and Suaib Luqman. "Narirutin. A flavonoid found in citrus fruits modulates cell cycle phases and inhibits the proliferation of hormone- refractory prostate cancer cells by targeting hyaluronidase." *Food and Chemical Toxicology* vol 174, (2023): 113638. https://doi.org/10.1016/j.fct.2023.113638 (**IF:4.3**)
6. **Akhilesh Kumar Maurya**, Nidhi Mishra, Coumarin-based combined computational study to design novel drugs against Candida albicans. *J Microbiology*. 60, 1201–1207 (2022) https://doi.org/10.1007/s12275-022-2279-5. (IF: 3.4)
7. **Akhilesh Kumar Maurya** & Nidhi Mishra, In silico validation of coumarin derivatives as potential inhibitors against Main Protease, NSP10/NSP16-Methyltransferase, Phosphatase and Endoribonuclease of SARS CoV-2, *Journal of Biomolecular Structure and Dynamics*, (2020), vol-39, 18, 7306-7321. https://doi.org/10.1080/07391102.2020.1808075. (IF 4.4)
8. Shristi Modanwal, **Akhilesh Kumar Maurya**, Sourav Mishra, and Nidhi Mishra, Development of QSAR model using machine learning and molecular docking study of polyphenol derivatives against obesity as pancreatic lipase inhibitor." *Journal of Biomolecular Structure and Dynamics* Volume 41, 2023 - Issue 14, Pages 6569-6580 https://doi.org/10.1080/07391102.2022.2109753 **(IF 4.4)**.
9. **Akhilesh Kumar Maurya**, Viswajit Mulpuru, and Nidhi Mishra, Discovery of Novel Coumarin Analogs against the α-Glucosidase Protein Target of Diabetes Mellitus: Pharmacophore-Based QSAR, Docking, and Molecular Dynamics Simulation Studies, *ACS Omega* 2020 5 (50), 32234-32249, https://doi.org/10.1021/acsomega.0c03871. **(IF 4.2).**
10. Santosh K Maurya, Fatma Huma, **Akhilesh Kumar Maurya**, Nidhi Mishra, Hifzur R. Siddique. Role of lupeol in chemosensitizing therapy-resistant prostate cancer cells by targeting MYC, β-catenin and c- FLIP: in silico and in vitro studies. In Silico Pharmacology. 2022 Dec;10(1):1-6. https://doi.org/10.1007/s40203-022-00131-3 .
11. Santosh K. Maurya, **Akhilesh Kumar Maurya**, Nidhi Mishra & Hifzur R. Siddique, Virtual screening, ADME/T, and binding free energy analysis of anti-viral, anti-protease, and anti-infectious compounds against NSP10/NSP16 methyltransferase and main protease of SARSCoV-2, *Journal of Receptor and Signal Transduction*. Vol 40, 6, 605-612. https://doi.org/10.1080/10799893.2020.1772298. (IF 2.1).

12. Shagun Varshney, **Akhilesh Kumar Maurya**, Akash Kanaujia and Nidhi Mishra, Silica from waste rice husk and its polymeric hydrogel-based composite filler: synthesis, characterisation, and antibacterial activity, Advances In Materials And Processing Technologies, https://doi.org/10.1080/2374068X.2023.2193448. Volume 10, 2024 - Issue 3, Pages 1485-1500, **(IF=2.1)**
13. **Akhilesh Kumar Maurya**, Shagun Varshney, Nidhi Mishra. Synthesis and characterization of polymeric hydrogel-based nanoporous composite and investigation of its temperature-dependent drug release activity. Materials Physics and Mechanics. 2022, 50(2):275-286. https://doi.org/10.18149/MPM.5022022_8.
14. Azad, Iqbal, Tahmeena Khan, **Akhilesh Kumar Maurya**, Mohd Irfan Azad, Nidhi Mishra, and Amer M. Alanazi. "Identification of Severe Acute Respiratory Syndrome Coronavirus-2 inhibitors through in silico structure-based virtual screening and molecular interaction studies." *Journal of Molecular Recognition* (2021), Vol 34,(10), e2918. https://doi.org/10.1002/jmr.2918, (IF 2.89)

Scopus Journals

1. Shagun Varshney, Ram Pravesh Pandey, Shivam Mishra **Akhilesh Kumar Maurya**, Nidhi Mishra; Biogenic synthesis of silver nanoparticles using Azadirachta indica and Thevetia peruviana leaf extracts: Characterization, antibacterial analysis, machine learning-based toxicity, and In-Silico molecular docking studies for potential biomedical applications, Nano-Structures & Nano-Objects, Volume 45, 2026, 101592, https://doi.org/10.1016/j.nanoso.2025.101592.
2. **Akhilesh Kumar Maurya**, Varshney, S., Upadhyay, S., Gupta, A., Jeyakumar, E., Lawrence, R., & Mishra, N. (2024). Green synthesis of copper oxide nanoparticles using Ficus racemosa leaf extract and investigation of its antibacterial properties. Advances in Materials and Processing Technologies, 1–21. https://doi.org/10.1080/2374068X.2024.2402970, (IF=2.1).
3. **Akhilesh Kumar Maurya**, Shagun Varshney, Vinod Verma, Hifzur R. Siddique & Nidhi Mishra, Synthesis of Silver and Copper oxide nanoparticles using Ficus racemosa leaf extract: characterization, anticancer potential, and dye degradation efficacy, Journal of Umm Al-Qura University for Applied Sciences. (2024). https://doi.org/10.1007/s43994-024-00182-6.
4. **Akhilesh Kumar Maurya**, Akshi Gupta, Rubina Lawrence, Nidhi Mishra: Microwave Assisted Green Synthesis of Silver using Ficus racemosa and their antibacterial and catalytic application in dye degradation, Research Journal of Chemistry and Environment 2021, 25(4):1-9. (IF 0.4) 0972-0626.
5. **Akhilesh Kumar Maurya**, Akshi Gupta, Rubina Lawrence, Nidhi Mishra: Microwave Assisted Green Synthesis of Silver using Ficus racemosa and their antibacterial and catalytic application in dye degradation, Research Journal of Chemistry and Environment 2021, 25(4):1-9. (IF 0.4) 0972-0626.

Conference Publications

1. **Akhilesh Kumar Maurya**, Nidhi Mishra. In Silico Study of Coumarin Derivatives Against Enoyl ACP Reductase, Ornithine Acetyltransferase and Protein Kinase B Target Enzymes of Mycobacterium tuberculosis. In 2019 IEEE 16th *India Council International Conference (INDICON)* 2019 Dec 13 (pp. 1-4). IEEE.
2. Viswajit Mulpuru, **Akhilesh Kumar Maurya**, Nidhi Mishra. CFD Analysis of Anti cancerous Ag and CuO Nanoparticles in Tumor Angiogenic Vessels. In2019 IEEE

16th *India Council International Conference (INDICON)* 2019 Dec 13 (pp. 1-2). IEEE.

Book Chapters

- Methods in Drug Repurposing: Emphasis on COVID-19 (Bentham Science)
- Polymeric Biomaterials in Tissue Engineering (CRC Press, Taylor & Francis)
- Smart Polymeric Biomaterials in Tissue Engineering (Apple Academic Press)
- Nanoengineered Polymeric Biomaterials for Drug Delivery (Elsevier)

Fellowships & Awards

- European Society of Medical Oncology (ESMO) Travel Grant (£1400) – EMSO-TAT-2025, Paris
- Gold Medal – Young Scientist Award (IAZ Conference 2021)
- Junior Research Fellowship – IIIT Allahabad
- CSIR NET Qualified (Life Sciences), June 2019
- GATE (Life Sciences) 2017 & 2019
- NSS Service (2012–2014)

Conference Presentations

1. **Akhilesh Kumar Maurya,** Discovery of PI3Kα Inhibitors for Breast Cancer through Drug Similarity, ESMO Targeted Anticancer Therapy 2025, organized by ESMO, Paris, France, 3rd – 5th March 2025
2. **Akhilesh Kumar Maurya,** Discovery of PI3Kα Inhibitors for Breast Cancer through Drug Similarity, CTDDR-2025, organized by CSRI-CDRI, Lucknow 19th – 22nd February 2025.
3. **Akhilesh Kumar Maurya,** Discovery of PI3Kα Inhibitors for Breast Cancer through Drug Similarity, Pan- IIT Meeting and Conference on Engineering in Medicine 2024, organized by IIT Kanpur, 6th – 8th December 2024
4. **Akhilesh Kumar Maurya,** Discovery of PI3Kα Inhibitors for Breast Cancer through Drug Similarity Analysis, **Poster Presentatio**n, SFRR-INDIA-DISCOVER-2024, BARC, Mumbai- India, Nov 6-9 2024.
5. **Akhilesh Kumar Maurya**, "Novel drug discovery against breast and lung cancer using pharmacophoric model based coumarin derivative: An in-silico Approach." **Poster Presentation**, International Conference on Drug Discovery 2022, Jointly Organized by Schrödinger and BITS Pilani K. K. Birla Goa Campus, 10-11 Nov 2022.
6. **Akhilesh Kumar Maurya** "Designing of 3D-QSAR Model of coumarin derivatives

against *Candida albicans*: An *in-silico* Approach", **Oral Presentation**, 5th Annual Meeting of International Association of Zoologists and 2nd International Virtual Seminar on Recent Trends in Life Sciences and Biotechnologist, organized by International Association of Zoologist (IAZ) December 12th -18th, 2021.

7. **Akhilesh Kumar Maurya**, In silico study of Coumarin derivatives Against PPR-Gamma and PPT-beta target protein diabetic type-2, **Poster Presentation**, Global Meet on Pharmaceutical Science, Drug Formulation and Delivery, Jacobs Conferences, Jun 19-20 2019. Venice, Italy
8. **Akhilesh Kumar Maurya** "In Silico Study of Coumarin Derivatives Against Enoyl ACP Reductase,
Ornithine Acetyltransferase and Protein Kinase B Target Enzymes of Mycobacterium tuberculosis. **Oral presentation**, 16th India Council International Conference (INDICON) 2019, organized by IEE Gujrat Section, December 13-15 2019

Training Programs & Workshops

- AICTE ATAL FDP (Healthcare AI, IIIT Allahabad, 2025)
- AICTE ATAL FDP (Healthcare & MedTech, DTU, 2024)
- KARYASHALA Hands-on Cell Culture Training (BHU, 2022)
- Molecular Biology Training (CytoGene R&D, 2022)
- NGS & CADD Workshop (IDAC, University of Lucknow, 2018)

Professional Memberships

- Member, European Society of Medical Oncology (ESMO)
- Life Member, Society for Free Radical Research International (SFRR)
- Life Member, International Association of Zoologists (IAZ)
- Life Member, Indian Society of Cell Biology (ISCB)
- Editorial Board Member, The Open Medicine Journal (Bentham Science)
- Peer Reviewer for scientific journals

Teaching Experience

- B.Sc. Biotechnology: Plant Physiology, First Aid and Health
- M.Sc. Biotechnology: Food Science and Technology
- Teaching Assistant, IIIT Allahabad (2018–2022)
Courses: Cheminformatics (PG), Molecular Medicine (PG)
- Guided PG projects and internships in CADD and Nanoparticle Synthesis

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Invited Lectures

- Drug Design and Development Against Breast Cancer – IIT Bombay
- Discovery of New Therapeutics via Computational Biology – Kalinga University (2024)

Referees

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